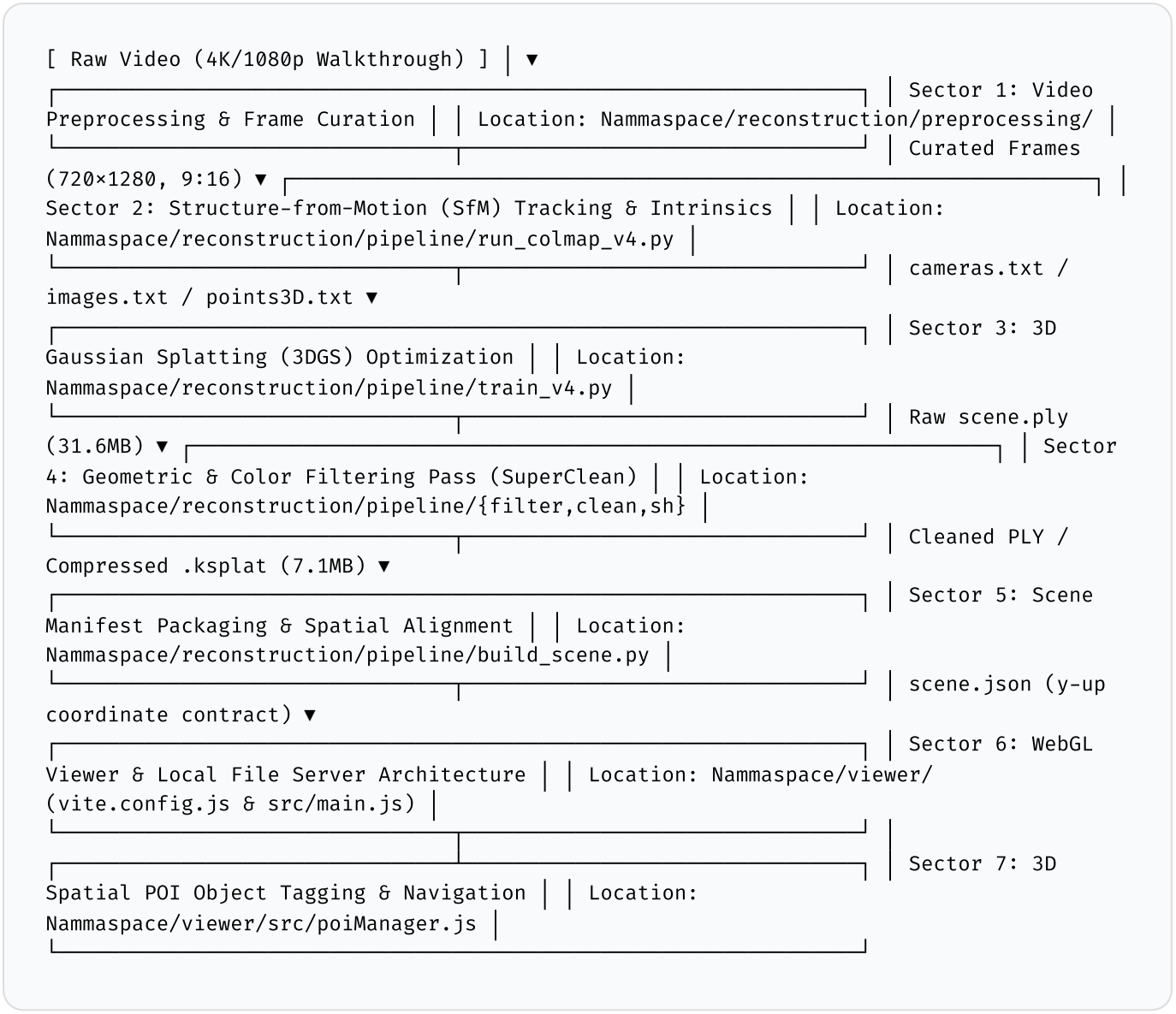
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Complete Technical Reference & Pipeline Specification across 7 Operational Sectors

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Sector 1: Video Preprocessing & Frame Curation

`reconstruction/preprocessing/extract_curate_v4.py`

PRIMARY TOOLS	INPUT SOURCE
FFmpeg 9.0.2, OpenCV (Laplacian), NumPy	IMG_3780.MOV (4K 120FPS HDR, iPhone 17 Pro Max)
OUTPUT TARGET	STANDARD RESOLUTION
data/processed/custom-video-v4/images/ (376 frames)	720 × 1280 px (9:16 portrait upright)

Technical Responsibilities:

- **Resolution & Rotation Normalization:** Honors the iPhone sensor rotation (-90°) while downscaling native 4K to 720×1280 to prevent CPU/GPU memory thrashing.
- **Laplacian Variance Filtering:** Samples candidates at 6 FPS into temporal buckets of 2.5 FPS, scoring spatial gradients via $\text{Var}(\nabla^2 I)$ to automatically drop motion-blurred frames.

```
def run_pipeline(video_path, base_out_dir, target_fps=2.5, sample_fps=6.0, max_width=720
    # Standardize scaling while maintaining native upright aspect ratio
    vf_filters = [f"fps={sample_fps}"]
    if max_width: vf_filters.append(f"scale='min(iw,{max_width})':-2")

    # Step 1: Candidate frame extraction
    cmd = [ffmpeg_bin, "-y", "-i", video_path, "-vf", ",".join(vf_filters), "-qscale:v",
    subprocess.check_call(cmd)

    # Step 2: Temporal window curation via Laplacian variance
    for bucket in time_windows:
        best_cand = max(bucket, key=lambda p: cv2.Laplacian(cv2.cvtColor(cv2.imread(p), 
        shutil.copy2(best_cand, target_img_dir)
```

Sector 2: Structure-from-Motion (SfM) Tracking & Intrinsics

`reconstruction/pipeline/run_colmap_v4.py`

PRIMARY TOOL	CAMERA MODEL
COLMAP 4.2.0 (14 CPU Worker Threads)	OPENCV (fx=1059.4px, fy=1050.3px, ~62° FOV)
REGISTERED VIEWS	3D ANCHOR POINTS
205 Cameras in Dominant Cluster	49,686 Triangulated Sparse Landmarks

Technical Responsibilities:

- **Multi-threaded SIFT:** Extracts ~6,500 scale-invariant features per frame across 14 CPU threads.
- **Sequential Overlap Matching:** Applies quadratic temporal lookahead (overlap=20) to track camera movement along the room walkthrough.
- **Non-Linear Bundle Adjustment:** Solves radial/tangential lens distortion (k1, k2, p1, p2) and establishes metric 3D camera trajectory.

```
# 1. Feature extraction
colmap feature_extractor --database_path db.db --image_path images/ --ImageReader.camera_

# 2. Sequential matching with quadratic overlap
colmap sequential_matcher --database_path db.db --SequentialMatching.overlap 20 --Sequent

# 3. Incremental mapping
colmap mapper --database_path db.db --image_path images/ --output_path sparse/ --Mapper.r
```

Sector 3: 3D Gaussian Splatting (3DGS) Training

`reconstruction/pipeline/train_v4.py`

PRIMARY ENGINE	ITERATIONS & STEPS
Brush 0.3.0 (CUDA 13.2 / RTX 4050 6GB)	15,000 Total Steps (Growth Freeze at 6,000)
SPHERICAL HARMONICS	TRAINING SPEED
SH Degree 2 (View-dependent radiance)	795.8 seconds (~13.2 minutes @ 100% GPU)

Technical Responsibilities:

- **Differentiable Splatting:** Projects 3D Gaussian ellipsoids to camera tiles, backpropagating L1 + SSIM loss.
- **Adaptive Density Splitting:** Clones and densifies under-represented surfaces up to step 6,000, freezing geometry thereafter for fine color optimization.

```
brush_app.exe Nammaspace/data/processed/custom-video-v4 \
    --total-steps 15000 \
    --growth-stop-iter 6000 \
    --growth-grad-threshold 0.00004 \
    --max-splats 1000000 \
    --sh-degree 2 \
    --export-path data/processed/custom-video-v4/train_v4 --export-name scene.ply
```

Sector 4: Geometric & Color Filtering Pass (SuperClean)

`reconstruction/pipeline/sh_clamp_filter.py`

NEEDLE CLAMPING

Aspect Ratio Capped at ≤ 4.0× (Eliminates Spikes)

FLOATER SUPPRESSION

SciPy cKDTree k-NN Distance Threshold (k=16)

SH DEFECT CLAMP

43,133 Splats (24.2%) Clamped from Negative RGB

RADIUS EXPANSION

1.8× Radial Scale (Bridges Gaps on Walls/Desk)

Technical Responsibilities:

- **Needle Clamping:** Eliminates razor-thin elongated Gaussians (up to 170:1) created by ray errors, maintaining clean planar shapes.
- **SOR Floater Filtering:** Removes isolated stray noise clusters floating in room air.
- **Spherical Harmonics Clamping:** Evaluates radiance from 35 camera viewpoints, zeroing overfitted Degree-2 rest coefficients for splats that drop below zero RGB to prevent black burn artifacts.

```
# Clamp extreme needle aspect ratios
if s_max > 4.0 * s_mid: scales[i, order[2]] = 4.0 * s_mid
if s_mid > 4.0 * s_min: scales[i, order[1]] = 4.0 * s_min
data[:, sc_cols] = np.log(np.maximum(scales * 1.8, 1e-8))

# Clamp view-dependent SH color burn
bad_splats = np.sum(np.any(total_rgb <= 0.05, axis=-1), axis=0) > 1
for sh2_idx in range(9, 24):
    data[bad_splats, idx[f"f_rest_{sh2_idx}"]] = 0.0
```

Sector 5: Scene Manifest Packaging & Spatial Alignment

`reconstruction/pipeline/build_scene.py`

UP-VECTOR FIX

v_world = -col_1(R_w2c) via Transposed Column 1

ALIGNMENT QUAT

[0.803322, 0.0, -0.581356, 0.129227] (Three.js y-up)

COMPRESSED KSPLAT

7.1 MB (168,637 Quantized Splats)

SCENE MANIFEST

viewer/public/scenes/custom-video-v4/scene.json

Technical Responsibilities:

- **Rodrigues Orientation Realignment:** Computes true camera-up gravity vector, solving the 90° sideways tilt.
- **Binary Quantization:** Converts raw 32-bit float PLY to compressed 7.1MB ksplat chunks for instant web delivery.

```
def _compute_cam_up(image_entry):
    r_cam2world = quat_to_rotmat(image_entry.qvec)
    # Correct transposed column 1 representation
    return (-r_cam2world[0][1], -r_cam2world[1][1], -r_cam2world[2][1])

# Quantize to ksplat chunk format
node viewer/convert-ksplat.mjs scene_clean.ply scene.ksplat 1 1 15
```

Sector 6: WebGL Viewer & Local File Server Architecture

`viewer/vite.config.js & viewer/src/main.js`

FILE SERVER

Vite 5.4 Dev Server (port 5173, host 0.0.0.0)

SECURITY HEADERS

COOP: same-origin | COEP: require-corp

SPLAT ENGINE

@mkkellogg/gaussian-splats-3d (DropInViewer)

RUNTIME FPS

60 – 165 FPS with Multi-Worker Radix Sort

Technical Responsibilities:

- **SharedArrayBuffer Isolation:** Server injects COOP/COEP headers allowing web workers to sort 160,000+ splats on CPU/GPU without blocking the UI thread.
- **Calibrated Spawn Target:** Places the camera directly inside the room viewing the desk, bed, and wardrobe.

```
// viewer/vite.config.js
export default defineConfig({
  server: {
    host: true,
    port: 5173,
    headers: {
      'Cross-Origin-Opener-Policy': 'same-origin',
      'Cross-Origin-Embedder-Policy': 'require-corp',
    },
  },
});
```

Sector 7: 3D Spatial POI Object Tagging & Navigation

`viewer/src/poiManager.js`

SCREEN PROJECTION	CAMERA FLIGHT
Three.js Matrix4 View-Projection Unproject	Ease-Out Cubic Camera Translation (1.2s duration)
VERIFIED POIS	PIPELINE GENERATOR
24 Indoor Objects (Almirahs, Desk, Bottles, Bed)	scripts/poi/poi_pipeline.py (YOLO-World + COLMAP)

Technical Responsibilities:

- **Geometric Label Anchoring:** Maps 3D world coordinates of furniture to 2D screen DOM pills without drift during camera rotation.
- **Interactive Camera Flight:** Clicking any POI triggers a smooth cinematic flight landing 0.8m directly in front of the object.

```
export class POIManager {
  update() {
    this.labels.forEach((item) => {
      const pos3D = item.worldPos.clone().project(this.camera);
      if (pos3D.z < 1.0) {
        const x = (pos3D.x * 0.5 + 0.5) * window.innerWidth;
        const y = (-pos3D.y * 0.5 + 0.5) * window.innerHeight;
        item.element.style.transform = `translate(-50%, -50%) translate3d(${x}px,${y}px,
      }
    });
  }
}
```